

CIS 5357

Computing for Data Analytics

JANUARY 18, 2024

Week 1

Chat with your neighbor

1. Introduce yourself
2. What is your coding/programming experience?
3. What are your personal goals for this course?
4. What are your biggest concerns regarding this course?

This week's goals

- Get introduced to the course and your classmates
- Install and get familiar with Python
- Run Python code from the command line/terminal

Data Analytics

the process of analyzing raw data in order to draw out meaningful, actionable insights, which are then used to inform and drive business decisions



Course goals

- use an **integrated development environment** (IDE) and **Jupyter** Notebook for creating computing applications;
- design and implement **decision and repetition structures** in Python;
- use Python collections and **libraries** in creating applications;
- implement **functions** in Python applications;
- test and **debug** Python applications;
- perform **basic data analysis** using Python libraries; and
- perform **data visualization** using Python libraries.



A **multiple-purpose** language with a **simple**, and **beginner-friendly** syntax.



Tools become obsolete but **your ability to learn new frameworks and solve problems** does not!

Syllabus overview

- Hybrid class format
- Class lectures & assignments will help synthesize info
- Note attendance policy; advance notice of absence is appreciated
- Get familiar with the honor code
- Schedule subject to change; Canvas will be the source of truth
- Read through syllabus and reach out if anything is unclear

Course assessments

Quizzes (2 @ 10% each)	20%
Assignments (4 @10% each)	40%
Exams (2 @ 20% each)	40%

No late assignments will be accepted & no make up exams—
plan your schedules accordingly

Course structure

PART 1: PYTHON BASICS

Weeks 1-7

Chapters 1-4

PART 2: DATA ANALYSIS

Weeks 8-16

Chapters 5-9



Download the textbook files

Module 1

Read and View

Read

- Textbook Introduction & Chapter 1 (pages 1-38). Section 1.3 is covered in the [Install Software](#) page of the Start Here module. Note the book mentions the text editor Atom, but we will use Visual Studio Code (or whatever your preferred text editor is).
- [Python: The New MBA Must-Have](#) ➞

For reference:

- [Command line cheatsheet](#) ➞
- [Python for MBAs files](#) ➞—download and save these files

View

[What is Python? Why Python is So Popular?](#) ➞



What is Python? Why Python is So Pop... Watch later Share

GitHub repo for code in textbook

The screenshot shows the GitHub repository page for `danguetta/python-for-mbas`. The repository is public and has 12 stars and 11 forks. The 'Code' button is highlighted with a red arrow. A dropdown menu is open, showing options to clone the repository using HTTPS or GitHub CLI, or to open it with GitHub Desktop. The 'Download ZIP' option is also visible. A second red arrow points to the 'Download ZIP' option. The repository contains files: `Part 1`, `Part 2`, `.gitattributes`, `.gitignore`, and `README.md`. The `README.md` file is open, showing the title 'Python for MBAs' and a description: 'This GitHub repository contains the code and data for "Python for MBAs", by Mattan Griffel and Daniel Guetta. The book is available from [Amazon.com](#) and [Columbia University Press](#).'

Download, unzip &
save code files

Open command line

Command line an application we can use to run Python code (among many other things)

macOS: Terminal

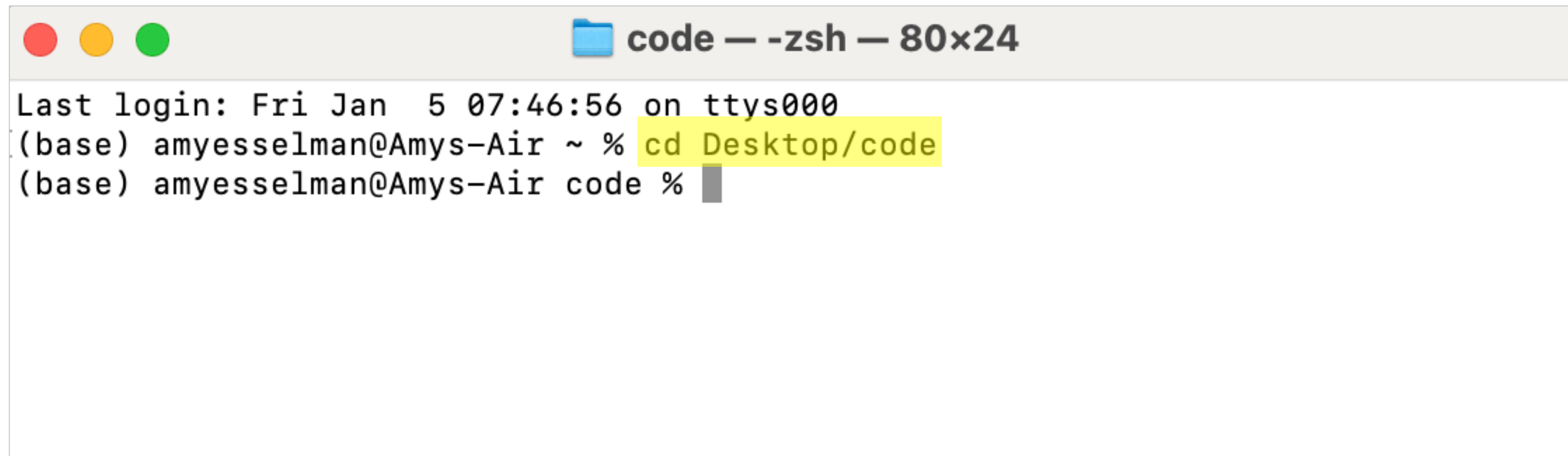
Click on the magnifying glass in the top, right-hand corner of your screen and type “Terminal”

Windows: Anaconda PowerShell Prompt

Click Start and type “Anaconda Powershell Prompt”



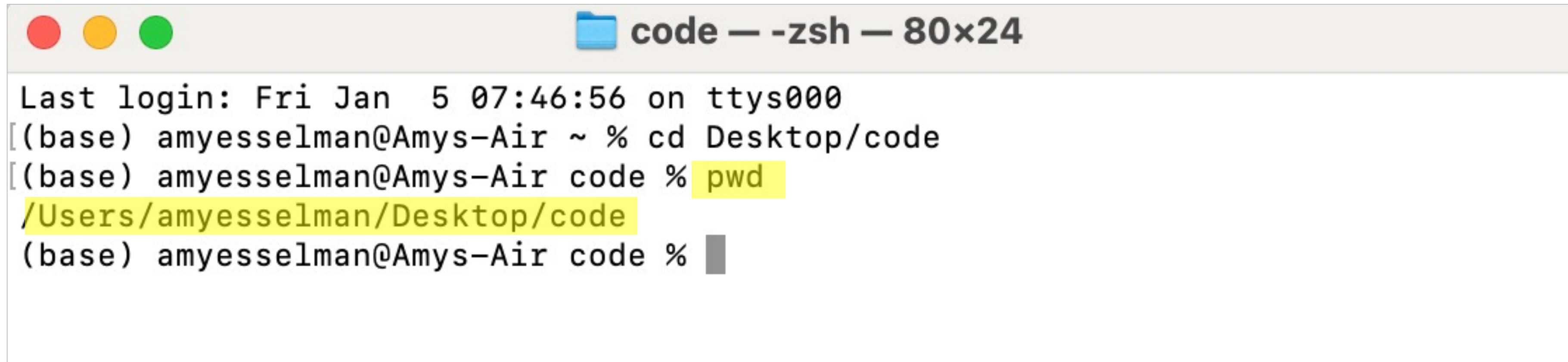
Change to directory with code

A terminal window titled "code — -zsh — 80x24" with a folder icon. The window shows the output of a login session: "Last login: Fri Jan 5 07:46:56 on ttys000". The prompt is "(base) amyesselman@Amys-Air ~ %". The command "cd Desktop/code" is entered and highlighted in yellow. The prompt changes to "(base) amyesselman@Amys-Air code %" with a cursor at the end.

```
Last login: Fri Jan 5 07:46:56 on ttys000
(base) amyesselman@Amys-Air ~ % cd Desktop/code
(base) amyesselman@Amys-Air code %
```

cd is the command to **change directories**
cd ~ will take you back to your **home directory**
cd .. will take you **up one level of folders**

Where am I?

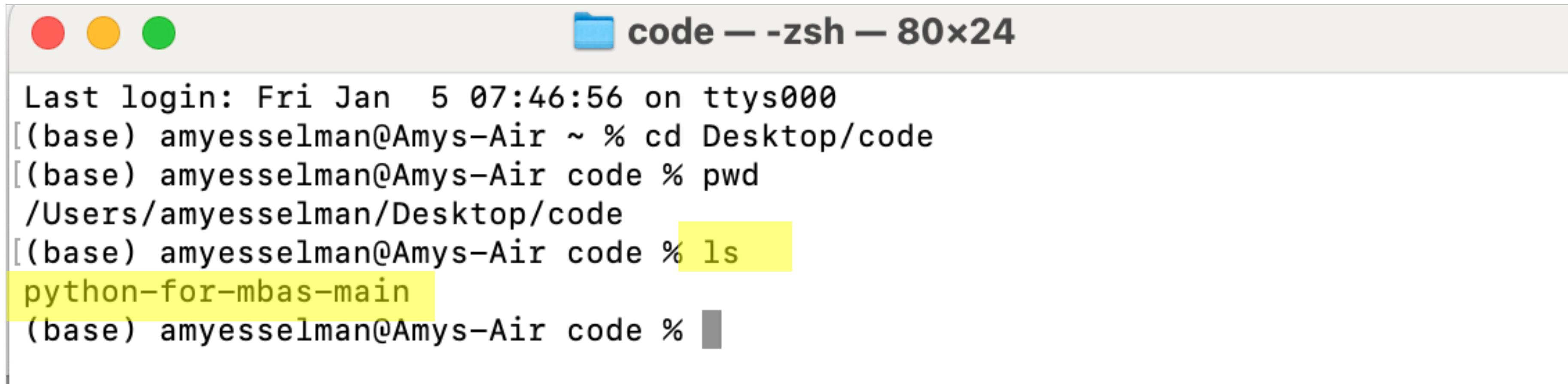
A screenshot of a terminal window titled "code — -zsh — 80x24". The window shows the following text: "Last login: Fri Jan 5 07:46:56 on ttys000", "[(base) amyesselman@Amys-Air ~ % cd Desktop/code]", "[(base) amyesselman@Amys-Air code % pwd]", and the output "/Users/amyesselman/Desktop/code" highlighted in yellow. The prompt "[(base) amyesselman@Amys-Air code %" is followed by a cursor.

```
Last login: Fri Jan 5 07:46:56 on ttys000
[(base) amyesselman@Amys-Air ~ % cd Desktop/code
[(base) amyesselman@Amys-Air code % pwd
/Users/amyesselman/Desktop/code
[(base) amyesselman@Amys-Air code %
```

pwd is the command to **print the working directory**

shortcuts: **TAB** autocompletes commands, functions, and filenames
UP ARROW accesses previous commands

List files using ls command



```
code — -zsh — 80x24
Last login: Fri Jan  5 07:46:56 on ttys000
[(base) amyesselman@Amys-Air ~ % cd Desktop/code
[(base) amyesselman@Amys-Air code % pwd
/Users/amyesselman/Desktop/code
[(base) amyesselman@Amys-Air code % ls
python-for-mbas-main
(base) amyesselman@Amys-Air code %
```

Move to Part 1 folder & view files

```
Part 1 — -zsh — 80x24

Last login: Fri Jan  5 07:48:06 on ttys000
[(base) amyesselman@Amys-Air ~ % cd Desktop/code
[(base) amyesselman@Amys-Air code % pwd
/Users/amyesselman/Desktop/code
[(base) amyesselman@Amys-Air code % ls
python-for-mbas-main
[(base) amyesselman@Amys-Air code % cd python-for-mbas-main/Part\ 1
[(base) amyesselman@Amys-Air Part 1 % ls
census.csv          lists.py
dictionaries.py      logic_practice.py
finance_functions.py logic_practice_solutions.py
fizzbuzz.py          loops.py
functions.py          math2.py
happy_hour.py         print.py
happy_hour2.py        strings.py
if.py                tip_calculator.py
importing.py          variables.py
input.py
(base) amyesselman@Amys-Air Part 1 %
```

If there is a space between words put a \ (backslash) before the space

WINDOWS: paths with spaces require quotes

Run happy_hour.py

```
Part 1 — -zsh — 80x24
Last login: Fri Jan  5 07:48:06 on ttys000
[(base) amyesselman@Amys-Air ~ % cd Desktop/code
[(base) amyesselman@Amys-Air code % pwd
/Users/amyesselman/Desktop/code
[(base) amyesselman@Amys-Air code % ls
python-for-mbas-main
[(base) amyesselman@Amys-Air code % cd python-for-mbas-main/Part\ 1
[(base) amyesselman@Amys-Air Part 1 % ls
census.csv          lists.py
dictionaries.py      logic_practice.py
finance_functions.py logic_practice_solutions.py
fizzbuzz.py          loops.py
functions.py         math2.py
happy_hour.py        print.py
happy_hour2.py       strings.py
if.py                tip_calculator.py
importing.py         variables.py
input.py
[(base) amyesselman@Amys-Air Part 1 % python happy_hour.py
How about you go to Death & Company with Sarah?
(base) amyesselman@Amys-Air Part 1 %
```

python is the command to **run Python code** (*.py files)

happy_hour.py

main [python-for-mbas](#) / [Part 1](#) / [happy_hour.py](#) / <> Jump to ▾



danguetta Initial commit

Lat

1 contributor

16 lines (12 sloc) | 376 Bytes

```
1 import random
2
3 bars = ["McSorley's Old Ale House",
4         "Death & Company",
5         "The Back Room",
6         "PDT"]
7
8 people = ["Mattan",
9           "Sarah",
10          "that person you forgot to text back",
11          "Samule L. Jackson"]
12
13 random_bar = random.choice(bars)
14 random_person = random.choice(people)
15
16 print(f"How about you go to {random_bar} with {random_person}?")
```

Challenge 1: Happy Hour Fixes

1. Oops, Samuel L. Jackson's name is spelled wrong. Can you fix it?
2. Add your favorite hangout spot to the bar list.
3. Have it print out two random bars instead of just one.

Challenge 1: One possible solution

```
1  import random
2
3  bars = ["McSorley's Old Ale House",
4          "Death & Company",
5          "The Back Room",
6          "PDT",
7          "Bangers"]
8
9  people = ["Mattan",
10           "Sarah",
11           "that person you forgot to text back",
12           "Samuel L. Jackson"]
13
14  random_bar = random.choice(bars)
15  random_bar2 = random.choice(bars)
16  random_person = random.choice(people)
17
18  print(f"How about you go to {random_bar} or {random_bar2} with {random_person}?")
```

Challenge 2: Consider edge cases

What if the *happy_hour.py* code randomly selects the same two places? How would you **conceptually get around this** problem? (You do not need to solve this in Python)

Some potential approaches

TWO DISTINCT LISTS: Create two mutually exclusive lists of bars and randomly select from each one. Since the lists have no overlap, there will be no duplication.

A LIST OF PAIRS: Create a list of all possible paired names (with no duplicates) and randomly select from that list.

LOGIC LOOP: After both random selections, check if they are same. If they are same, run the random selection again. Keep doing this until they do not match.

REMOVE FROM LIST: After the first random person gets selected, assign a new list that excludes the person selected and then randomly select one person from the resulting list.

Recap

- Importance of data analysis for business
- Course overview
- Python for data analytics
- Running a simple Python script

Next week

- Welcome quiz

Start your homework early. If you run into technical issues or have questions, you'll want to have time to get help!

- Topics for next week

- Data types
- Debugging code